

Arrays - VI

Wednesday, 22 April 2026

9:33 PM

Interview Q.

Sort an array of 0s, 1s and 2s.

arr = {1, 1, 0, 1, 2, 0, 1, 2}
 $\Rightarrow \{0, 0, 1, 1, 1, 1, 2, 2\}$

arr = {0, 2, 1, 0}

$\Rightarrow \{0, 0, 1, 2\}$

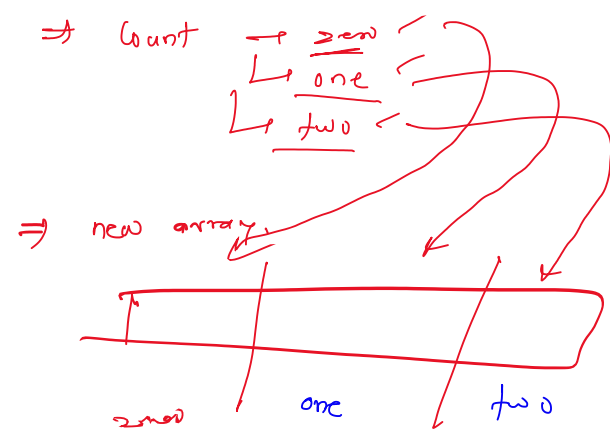
- Selection sort $\checkmark$ $\rightarrow$ TC $O(n^2)$
- bubble sort $\checkmark$ $\rightarrow$ $O(n^2)$
- insertion sort $\checkmark$ $\rightarrow$ $O(n^2)$
- Merge sort $\checkmark$ $\rightarrow$ $O(n \log n)$
- Quick sort $\checkmark$ $\rightarrow$ $O(n \log n)$

Approach 1 - Use sorting algo.

TC - $O(n^2)$

TC - $O(n \log n)$

Approach 2 - for disp.

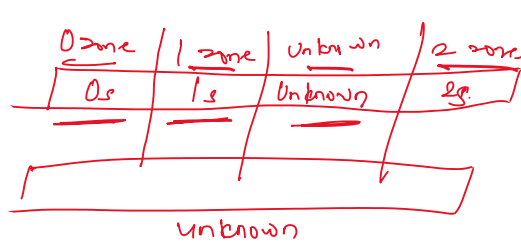


Approach 3 -

Algorithm $\rightarrow$ Dutch National Flag.

One Intuition-

Array $\Rightarrow$ 3 regions.



$\Rightarrow$ Starting, pura array $\rightarrow$ unknown

$\Rightarrow$ half half unknown aake track karke



low $\Rightarrow$ next 0 yeti to kutha jannar.

mid $\Rightarrow$ current element / decision maker.

high $\Rightarrow$ next 2 yeti to kutha jannar.

Decision kosa gheych?

Case (I) arr[mid] == 0

$\rightarrow$ swap (low) $\Rightarrow$

$\rightarrow$ 0-zone vadhaunna.

[0-zone | unknown]

[0-zone | unknown]

{ 2, 0, 2, 1, 1, 0 }

0 1 2 3 4 5

Case (II) . arr[mid] == 1

mid++;

Case (III) ~ arr[mid] == 2.

$\rightarrow$ swap (high)

$\rightarrow$ 2-zone expand.

Dry Run-

{ 2, 0, 2, 1, 1, 0 }

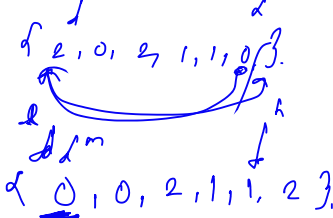
0 1 2 3 4 5

low $\Rightarrow$ 0

mid $\Rightarrow$ 0

high = 5

Step 1 - arr[mid] = 2.



Step 2 - arr[mid] = 0

{ 0, 0, 2, 1, 1, 2 }

{ 2, 0, 2, 1, 1, 2 }

Step 3 - arr[mid] = 0

{ 0, 0, 2, 1, 1, 2 }

{ 0, 0, 2, 1, 1, 2 }

Step 4 - arr[mid] = 2

{ 0, 0, 1, 1, 2, 2 }

{ 0, 0, 1, 1, 2, 2 }

Step 5 - arr[mid] = 1

{ 0, 0, 1, 1, 2, 2 }

Step 6 - arr[mid] = 1

{ 0, 0, 1, 1, 2, 2 }

{ 0, 0, 1, 1, 2, 2 }

Code -

int low = 0;

int mid = 0;

int high = arr.length - 1;

while (mid <= high) {

if (arr[mid] == 0) {

swap (arr, low, mid);

low++;

mid++;

} else if (arr[mid] == 1) {

mid++;

}

else {

swap (arr, mid, high);

high--;

}

Dry Run -

1) { 0, 2, 0, 0, 1, 1, 2, 0 }

